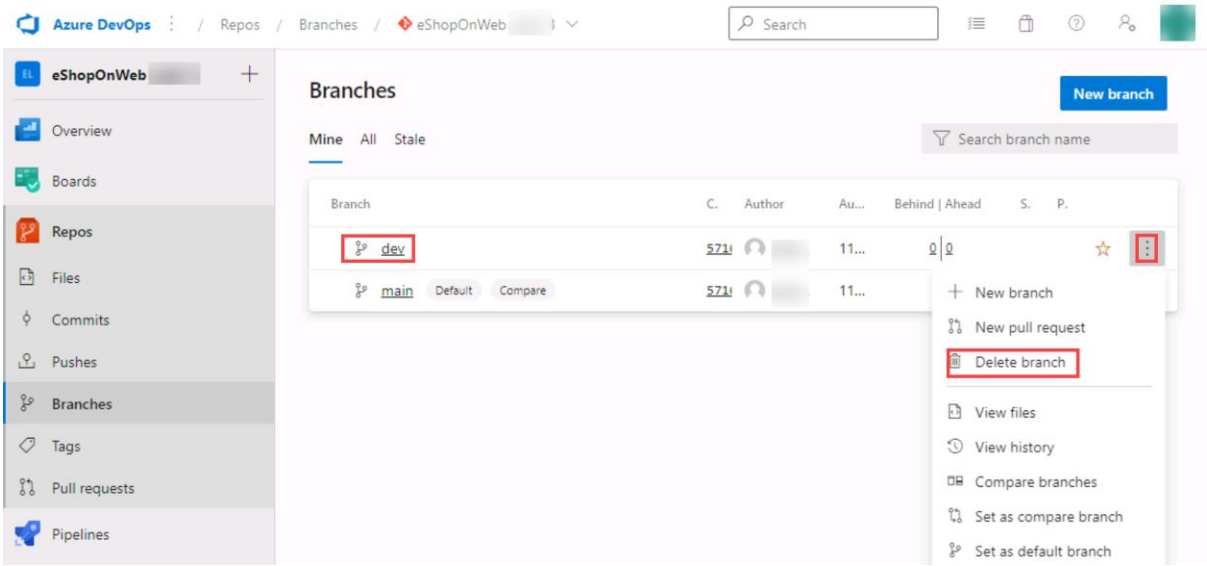
Delete a branch

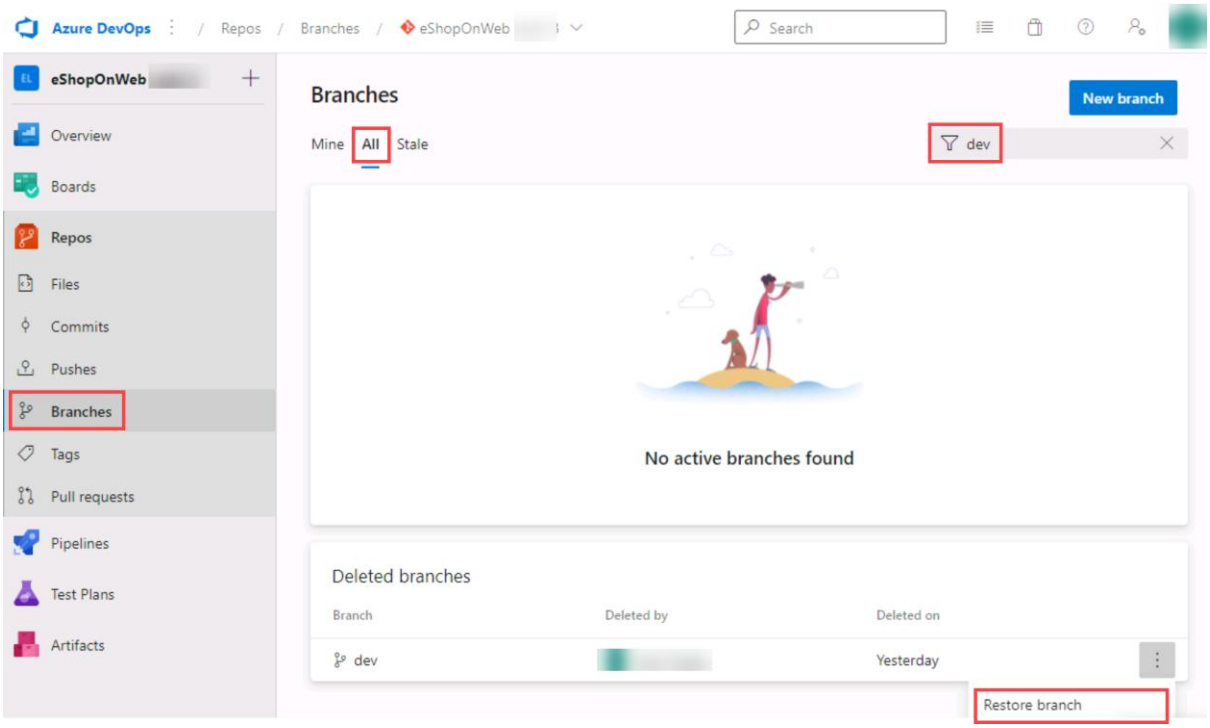
Git keeps track of which branch you are working on and makes sure that, when you check out a branch, your files match the most recent commit on that branch. Branches let you work with multiple versions of the source code in the same local Git repository at the same time. You can use Visual Studio Code to publish, check out and delete branches.

1. In the **Visual Studio Code** window, with the **SOURCE CONTROL** tab selected, in the lower left corner of the Visual Studio Code window, select the **Publish changes** icon (directly to the right of the **dev** label representing your newly created branch).
2. Switch to the web browser window displaying the **Commits** pane of the **Azure DevOps** portal and select **Branches**.
3. On the **Mine** tab of the **Branches** pane, verify that the list of branches includes **dev**.
4. Hover the mouse pointer over the **dev** branch entry to reveal the ellipsis symbol on the right side.
5. Select the ellipsis, in the pop-up menu, select **Delete branch**, and, when prompted for confirmation, select **Delete**.



6. Switch back to the **Visual Studio Code** window and, with the **SOURCE CONTROL** tab selected, in the lower left corner of the Visual Studio Code window, select the **dev** entry. This will display the existing branches in the upper portion of the Visual Studio Code window.
7. Verify that now there are two **dev** branches listed.

8. Go to the web browser displaying the **Mine** tab of the **Branches**
9. On the **Mine** tab of the **Branches** pane, select the **All** tab.
10. On the **All** tab of the **Branches** pane, in the **Search branch name** text box, type **dev**.
11. Review the **Deleted branches** section containing the entry representing the newly deleted branch.
12. In the **Deleted branches** section, hover the mouse pointer over the **dev** branch entry to reveal the ellipsis symbol on the right side.
13. Select the ellipsis, in the pop-up menu and select **Restore branch**.



Note: You can use this functionality to restore a deleted branch as long as you know its exact name.

Branch Policies

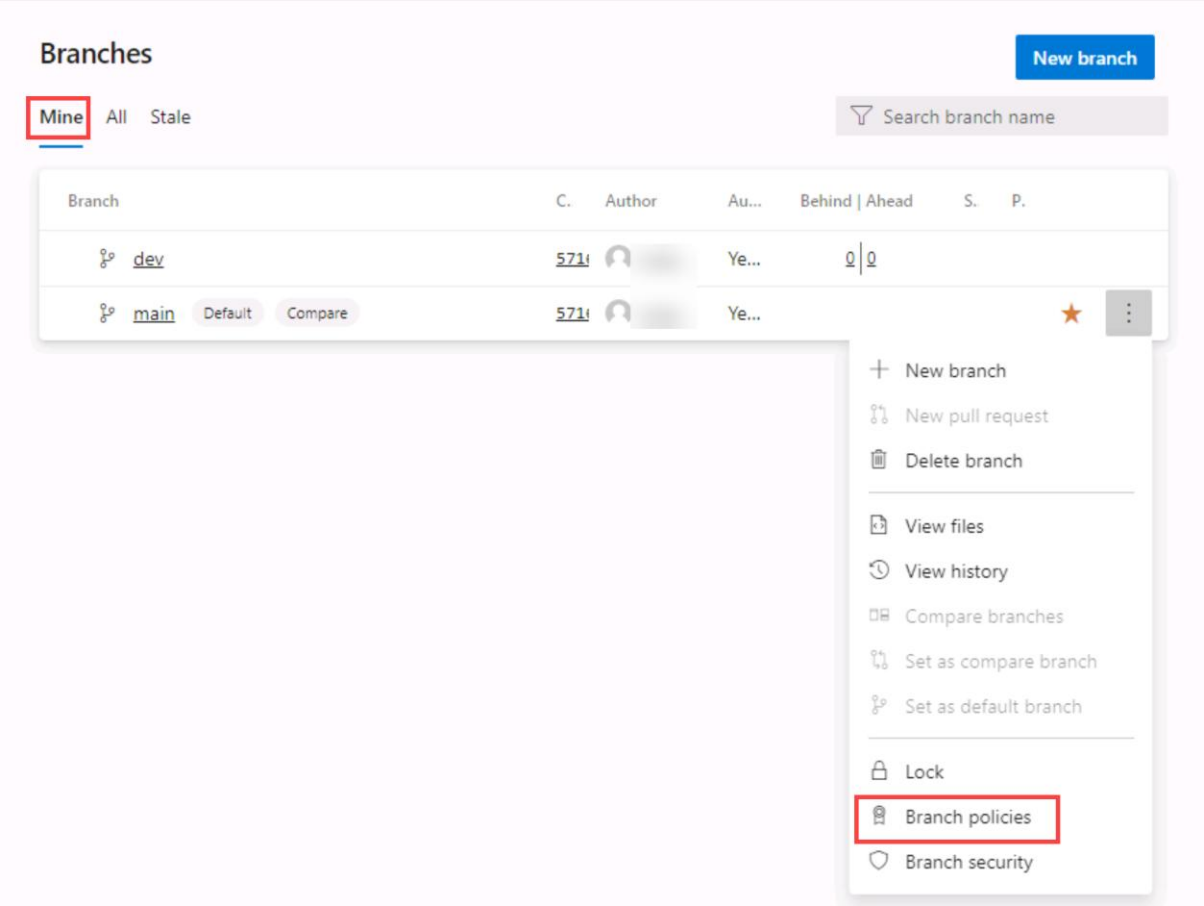
You will use the Azure DevOps portal to add policies to the main branch and only allow changes using Pull Requests that comply with the defined policies. You want to ensure that changes in a branch are reviewed before they are merged.

For simplicity we will work directly on the web browser repo editor (working directly in origin), instead of using the local clone in VS code (recommended for real scenarios).

1. Switch to the web browser displaying the **Mine** tab of the **Branches** pane in the Azure DevOps portal.

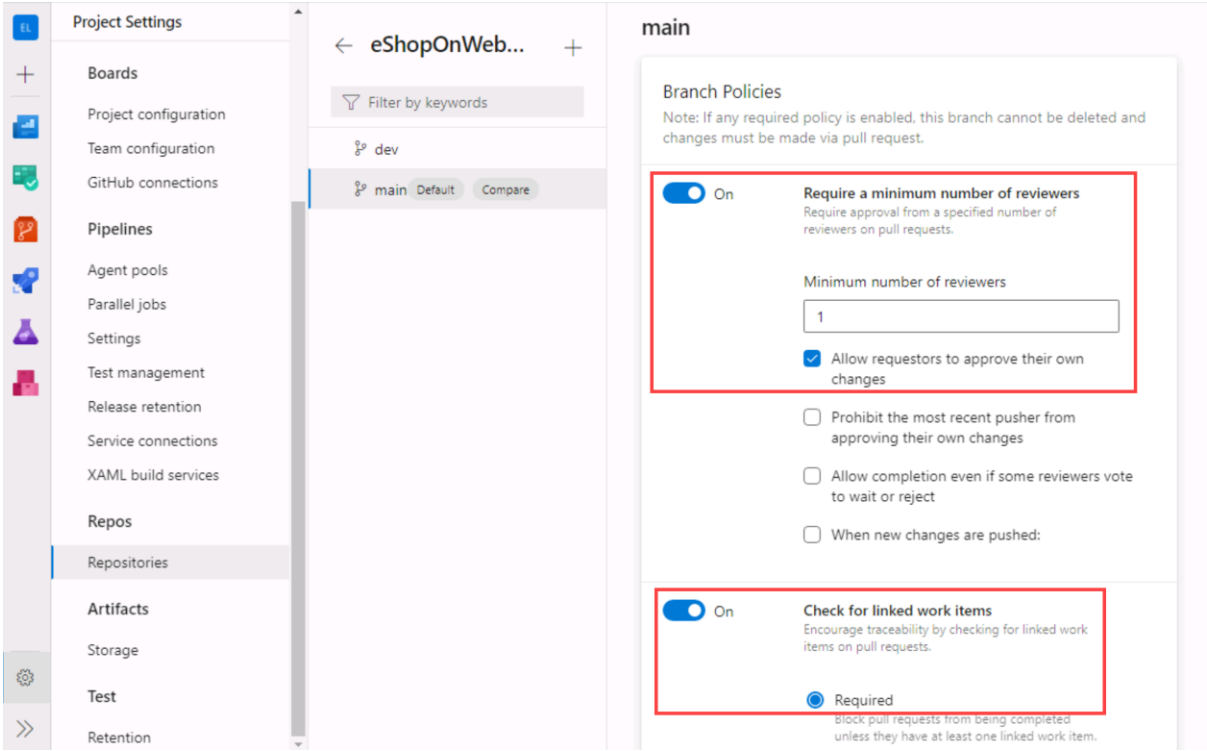
2. On the **Mine** tab of the **Branches** pane, hover the mouse pointer over the **main** branch entry to reveal the ellipsis symbol on the right side.

3. Select the ellipsis and, in the pop-up menu, select **Branch Policies**.



4. On the **main** tab of the repository settings, enable the option for **Require minimum number of reviewers**. Add **1** reviewer and check the box **Allow requestors to approve their own changes**(as you are the only user in your project for the lab)

5. On the **main** tab of the repository settings, enable the option for **Check for linked work items** and leave it with **Required** option.

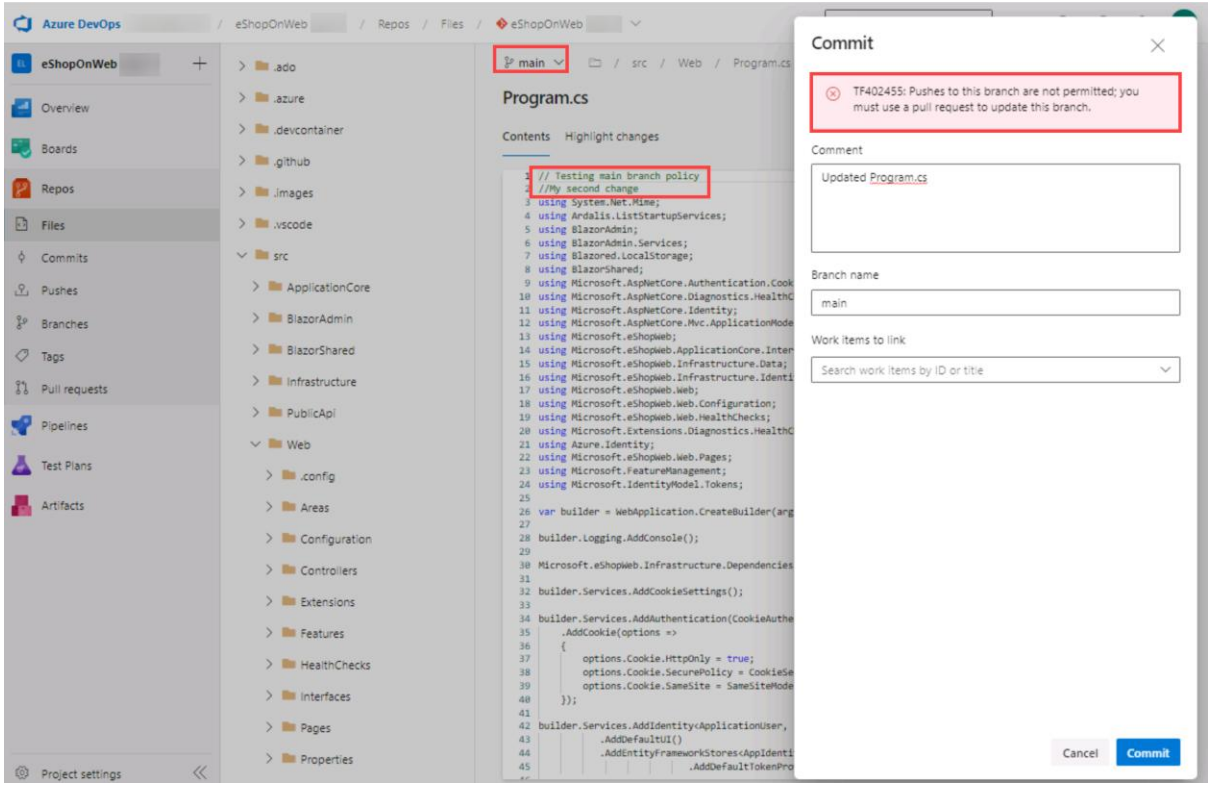


Testing branch policy

1. In the vertical navigational pane of the Azure DevOps portal, in the **Repos>Files**, make sure the **main** branch is selected (dropdown above shown content).
2. To make sure policies are working, try making a change and committing it on the **main** branch, navigate to the **/eShopOnWeb/src/Web/Program.cs** file and select it. This will automatically display its content in the details pane.
3. On the first line add the following comment:

// Testing main branch policy

4. Select **Commit > Commit**. You will see a warning: changes to the main branch can only be done using a Pull Request.



5. Select **Cancel** to skip the commit.

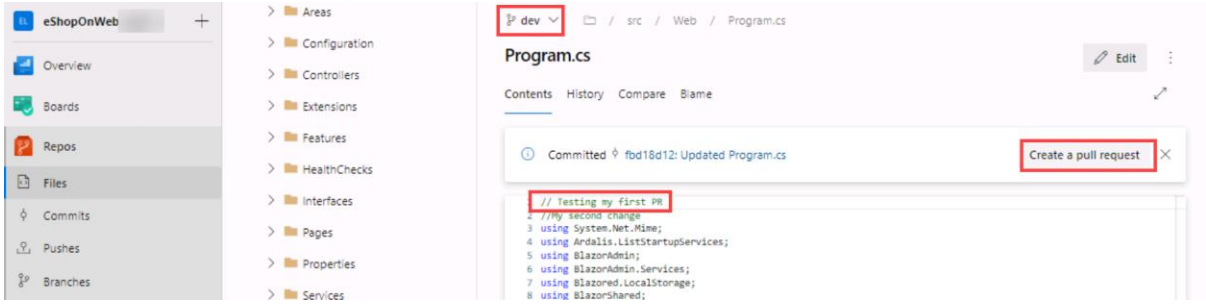
Working with Pull Requests

You will use the Azure DevOps portal to create a Pull Request, using the **dev** branch to merge a change into the protected **main** branch. An Azure DevOps work item will be linked to the changes to be able to trace pending work with code activity.

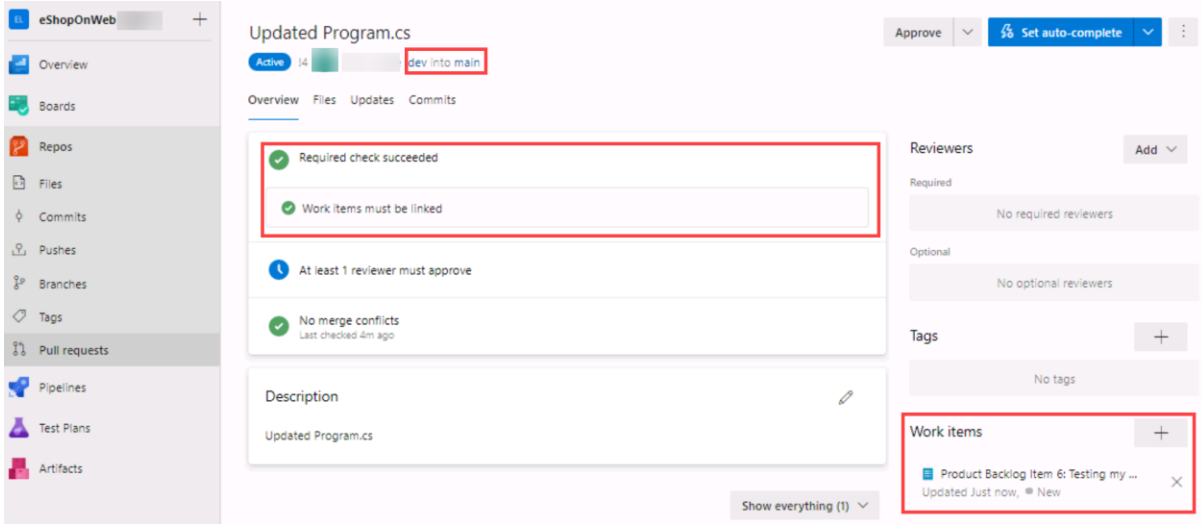
1. In the vertical navigational pane of the Azure DevOps portal, in the **Boards** section, select **Work Items**.
2. Select **+ New Work Item > Product Backlog Item**. In title field, write **Testing my first PR** and select **Save**.
3. Now go back to the vertical navigational pane of the Azure DevOps portal, in the **Repos>Files**, make sure the **dev** branch is selected.
4. Navigate to the **/eShopOnWeb/src/Web/Program.cs** file and make the following change on the first line:

// Testing my first PR

5. Select **Commit > Commit** (leave default commit message). This time the commit works, **dev** branch has no policies.
6. A message will pop-up, proposing to create a Pull Request (as you **dev** branch is now ahead in changes, compared to **main**). Select **Create a Pull Request**.



7. In the **New pull request** tab, leave defaults and select **Create**.
8. The Pull Request will show some failed/pending requirements, based on the policies applied to our target **main** branch.
    - Proposed changes should have a work item linked
    - At least 1 user should review and approve the changes.
9. On the right side options, select the **+** button next to **Work Items**. Link the previously created work item to the Pull Request by selecting on it. You will see one of the requirements changes status.



10. Next, open the **Files** tab to review the proposed changes. In a more complete Pull Request, you would be able to review files one by one (marked as reviewed) and open comments for lines that may not be clear (hovering the mouse over the line number gives you an option to post a comment).
11. Go back to the **Overview** tab, and on the top-right select **Approve**. All the requirements will change to green. Now you can select **Complete**.
12. On the **Complete Pull Request** tab, multiple options will be given before completing the merge:
    - **Merge Type**: 4 merge types are offered, you can review them here or observing the given animations. Choose **Merge (no fast forward)**.

- **Post-complete options**:
  - Check **Complete associated work item...**. It will move associated PBI to **Done** state.

13. Select **Complete Merge**

Applying tags

The product team has decided that the current version of the site should be released as v1.1.0-beta.

1. In the vertical navigational pane of the Azure DevOps portal, in the **Repos** section, select **Tags**.
2. In the **Tags** pane, select **New tag**.
3. In the **Create a tag** panel, in the **Name** text box, type **v1.1.0-beta**, in the **Based on** drop-down list leave the **main** entry selected, in the **Description** text box, type **Beta release v1.1.0** and select **Create**.

Note: You have now tagged the repository at this release (the latest commit gets linked to the tag). You could tag commits for a variety of reasons and Azure DevOps offers the flexibility to edit and delete them, as well as manage their permissions.

Remove Branch Policies

When going through the different course labs in the order they are presented, the branch policy configured during this lab will block exercises in future labs. Therefore, we want you to remove the configured branch policies.

1. From the Azure DevOps **eShopOnWeb** Project view, navigate to **Repos** and select **Branches**. Select the **Mine** tab of the **Branches** pane.
2. On the **Mine** tab of the **Branches** pane, hover the mouse pointer over the **main** branch entry to reveal the ellipsis symbol (the ...) on the right side.
3. Select the ellipsis and, in the pop-up menu, select **Branch Policies**.
4. On the **main** tab of the repository settings, disable the option for **Require minimum number of reviewers**.
5. On the **main** tab of the repository settings, disable the option for **Check for linked work items**.
6. You have now disabled/removed the branch policies for the main branch.

Clean up resources

You don't need to clean up your Azure DevOps organization or project, as they will remain available for you to use as a reference and portfolio item. Azure DevOps provides free tier usage that includes basic features for small teams.

If you want to delete the project, you can do so by following these steps:

1. In your browser navigate to the Azure DevOps portal at https:*//aex.dev.azure.com*.
2. Navigate to the **eShopOnWeb** project you created.
3. On the project settings page, go to **Overview** and select **Delete** at the bottom of the page.
4. Type the project name to confirm deletion and select **Delete**.

CAUTION: Deleting a project deletes all work items, repositories, builds, and other project artifacts. If you used an existing project for this exercise, any existing resources outside the scope of this exercise will also be deleted.